Unbinned simultaneous multi-observable unfolding of the MINERvA medium-energy CC-inclusive cross section

Joseph Bailey Benjamin Nachman

August 22, 2026

Abstract

We apply unbinned machine-learning unfolding (OmniFold) to MINERvA medium-energy ν_μ data. The unbinned $d^2\sigma/(dp_T\, dp_\parallel)$ reproduces the published iterative-Bayesian measurement (σ_{tot} ratio 1.011; median per-bin uncertainty 6.87 % vs 6.86 %). Without constructing a new explicit response tensor for each dimensional extension, we then perform *unbinned* simultaneous three-, four- and five-observable MINERvA unfolds, adding available energy E_{avail}, momentum transfer q_3 and hadronic mass W. The low-E_{avail} region independently recovers the known low-recoil/2p2h deficit and agrees at central-value level with the published low-recoil measurement without a corrected 4D covariance comparison; the (E_{avail}, W) plane localizes a residual deep-inelastic excess that all tested generators underpredict. A recoil-only point-cloud transformer supplies a method-development cross-check of a low-level representation; the muon is not a classifier input in that study. Removing the truth-level muon kinematic cuts also gives a preliminary extended-fiducial central value, $\sigma_{\mathrm{ext}} = 4.502 \times 10^{-38}\,\mathrm{cm}^2/\mathrm{nucleon}$, with acceptance-supported and prior-dominated regions reported separately.

1 Introduction

Differential cross sections from neutrino experiments are extracted by *unfolding* the detector response. The standard tool, iterative Bayesian unfolding on a binned migration matrix, scales poorly because each additional observable multiplies the bin count. Unbinned machine-learning unfolding (OmniFold [1, 2]) avoids constructing an explicit response tensor by learning per-event weights in the full observable space. Higher dimension still increases training, support, sparsity, and uncertainty-propagation demands.

We apply OmniFold to MINERvA medium-energy (ME) FHC ν_μ data. We first reproduce the published double-differential CC-inclusive cross section $d^2\sigma/(dp_T\, dp_\parallel)$ [3] as a validation, then extend the unbinned formulation to simultaneous three-, four- and five-observable MINERvA unfolds, adding available energy E_{avail}, three-momentum transfer q_3, and hadronic mass W. Binned MINERvA multi-differential cross sections, up to triple-differential [4], exist already; the methodological distinction is the unbinned, simultaneously-unfolded formulation rather than multi-dimensionality itself. These expose the known low-recoil/2p2h deficit through an independent route, localize a residual deep-inelastic excess that all four tested generators underpredict, and — together with a recoil-point-cloud representation cross-check and a preliminary extended-fiducial study — demonstrate the reach and current limitations of the method.

2 Data and method

The analysis uses the twelve ME FHC playlists (1.057×10^{21} protons on target, $\langle E_\nu \rangle \sim 6\,\mathrm{GeV}$), with the signal definition, fiducial volume, flux [5] and selection of Ref. [3]; after selection the data sample is 4.12M events against 32.8M simulated signal events (GENIE 2.12.6 with the MINERvA Tune v1). A dedicated event loop writes per-event truth and reconstructed observables for the unbinned unfold.

OmniFold unfolds by iterated reweighting. Each iteration trains a classifier to separate data from reconstructed simulation, converts its output to a per-event likelihood-ratio weight, pulls that weight back to truth through the simulation's truth–reco pairing (treating unmatched "miss" events natively, so no separate efficiency correction is applied), and trains a second classifier to propagate it to a smooth truth-level reweight for the next iteration. We use gradient-boosted decision trees (100 trees, depth-3 capacity, 5 iterations) as the learners: the published central value uses the `exact` backend (sklearn `GradientBoosting`, single-threaded exact-split CART), while the uncertainty ensemble — the systematic universe sweep and the Poisson bootstrap — uses LightGBM at matched capacity (`num_leaves` $= 8 = 2^3$), so the central value and its covariance are *not* estimator-matched. The higher-dimensional unfolds simply append feature columns $(p_T, p_{\|}, E_{\mathrm{avail}}, q_3, W)$. Histogramming the reweighted truth ensemble (in any declared reporting binning) and dividing by flux, exposure and target-nucleon count gives the cross section. For the finalized 2D result, the uncertainty budget combines systematic (MAT universe), statistical (Poisson bootstrap) and ML (training-seed variation) covariances; the complete statistical methodology and validation suite are documented in the companion analysis note.

3 Two-dimensional reproduction

The unbinned $d^2\sigma/(dp_T\, dp_{\|})$ reproduces the published measurement (Fig. 1): the total cross section is $3.073 \times 10^{-38}\,\mathrm{cm}^2$/nucleon (published $3.039 \times 10^{-38}\,\mathrm{cm}^2$/nucleon, ratio 1.011), 94.1 % of the 205 reported bins agree within 10 %, and the published-σ per-bin standardized differences are sub-1σ (RMS 0.598). Our standalone uncertainty budget matches the published total in size (median per-bin 6.87 % vs 6.86 %), flux-dominated as in the publication. The residual indicative paper-covariance distance of 3.66 per bin is shape, not normalization, and traces to the off-diagonal structure of the published covariance on the low-p_T peak; it carries a $\sim$ 1-unit method-dependence band from the choice of GBDT estimator (exact-GBT 3.66 versus LightGBM 2.65).

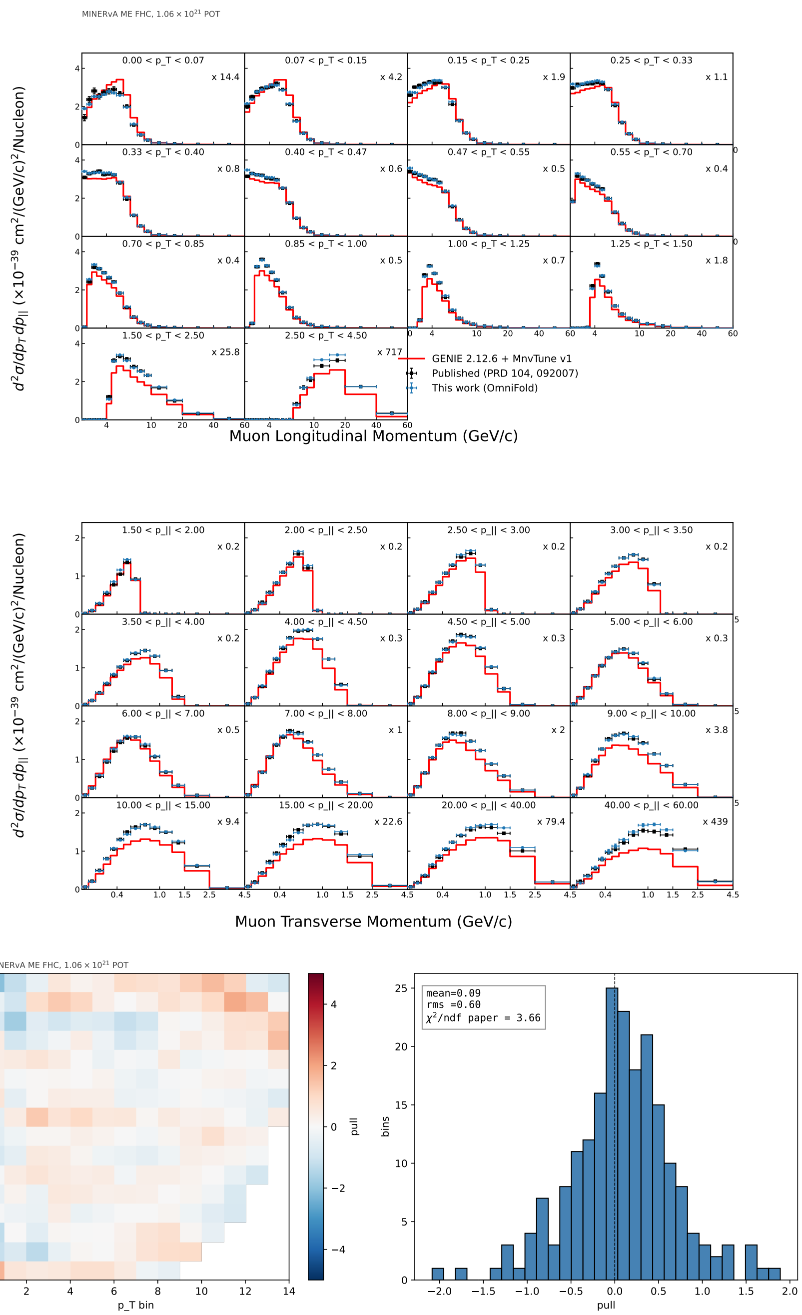

Figure 1: Top: unfolded 2D cross section. Bottom: per-bin standardized difference using the published uncertainty on the 14×16 grid; the distribution is sub-1σ (RMS 0.598). The two results share data, so this is descriptive rather than an independent-result pull.

4 Higher-dimensional cross sections

4.1 Available energy: the low-recoil deficit

Extending to $(p_T, p_{||}, E_{\text{avail}})$ and then $(p_T, p_{||}, E_{\text{avail}}, q_3)$ yields *unbinned* three- and four-observable MINERvA unfolds; each passes the projection anchors against its lower-dimensional predecessor to ≈1–2 %. Four truth-level generators (GENIE CV, MINERvA Tune v1, NuWro, GiBUU) all underpredict the data at low E_{avail} (Fig. 2), the established MINERvA low-recoil/2p2h deficit. We confirm its origin directly: pion final-state-interaction dials move $d\sigma/dE_{\text{avail}}$ by less than 1 %, whereas adding the empirical Valencia 2p2h fills the quasielastic–Δ dip. The four-dimensional (E_{avail}, q_3) result resolves the deficit's momentum-transfer structure, and a bin-identical comparison to the published low-recoil measurement on the maximal common grid [6] agrees at central-value level. No covariance-based consistency metric is reported without a corrected 4D covariance (Fig. 3).

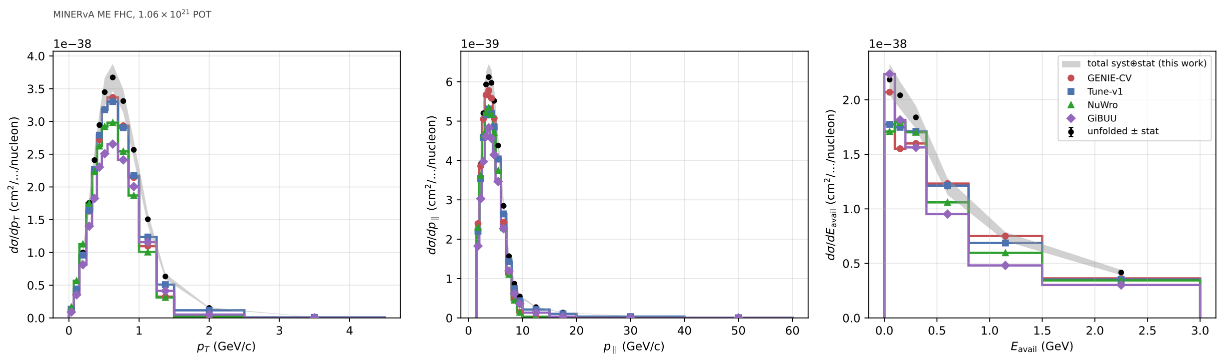

Figure 2: Unfolded 3D central cross section (black) versus four truth-level generators. The generators track the muon-momentum projections but split on E_{avail}, all underpredicting the low-recoil peak. Only the central values are interpreted.

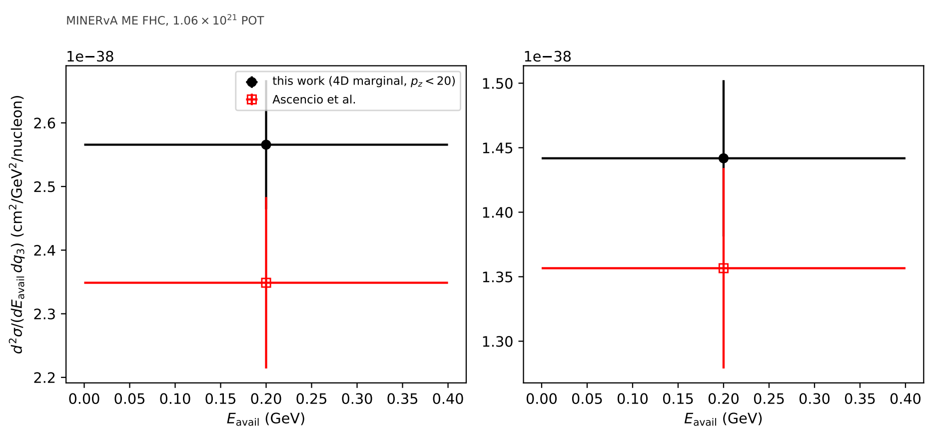

Figure 3: Bin-identical central-value comparison to the published low-recoil measurement [6] on the maximal common (E_{avail}, q_3) grid. The uncertainty-derived annotations do not use a corrected 4D covariance and are not interpreted.

4.2 Hadronic mass: a deep-inelastic excess no generator reproduces

Adding W as a fifth axis isolates the discrepancy that survives 2p2h. The unfolded-minus-generator excess concentrates at high E_{avail} *and* high W — the deep-inelastic region (Fig. 4). All four generators underpredict the central result in the high-W corner. Statistical significances are not reported without a corrected covariance for this projection. Enabling Valencia 2p2h does not close the corner — it slightly worsens it, because 2p2h populates low W — and the independent NuWro and GiBUU generators miss it by the same or a larger margin. The excess is thus a genuinely deep-inelastic ($W \gtrsim 1.8\,\mathrm{GeV}$) modeling deficit, robust across generator and tune.

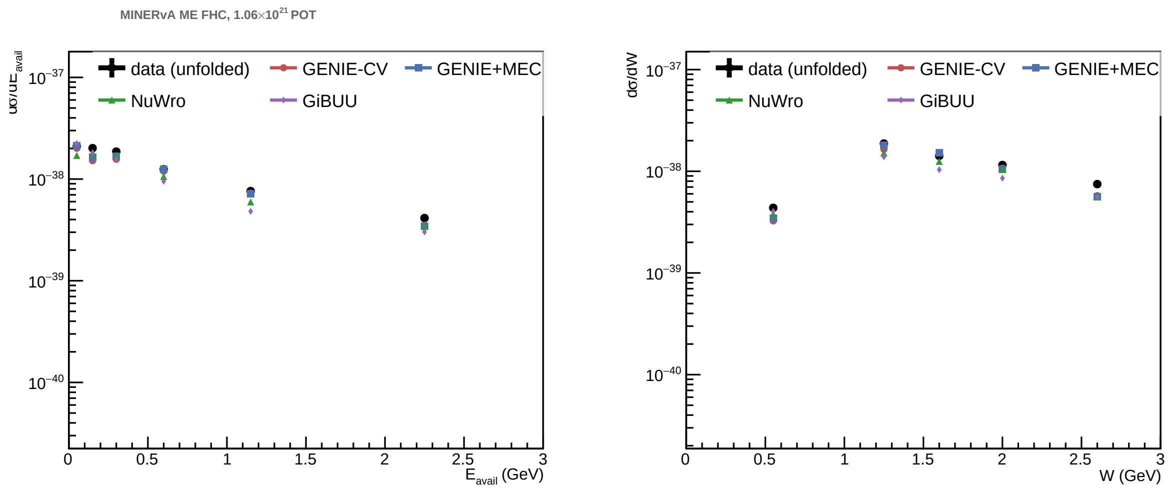

Figure 4: Unfolded (E_{avail}, W) cross section versus the four-generator band. Every generator underpredicts the high-W, high-E_{avail} deep-inelastic corner; 2p2h does not help there.

5 Method robustness and the recoil-point-cloud track

The unfolded result is stable across GBDT and MLP classifier families. A point-cloud transformer (PET [7]) provides a complementary recoil-representation cross-check: step 1 uses non-muon reconstructed calorimeter clusters and step 2 uses truth final-state hadrons with the muon removed. It yields an absolutely-normalized central value (Fig. 5). Its ordinary closure was assessed only as a legacy two-iteration self-consistency diagnostic, and no current closure level is quoted. This track is reported as a diagnostic, method-development study: it contributes no cross section and no uncertainty to the measurements above. No current quantitative PET–GBDT level of agreement is quoted. The only such comparison that exists was taken with a two-iteration training, against the campaign's pinned three-iteration policy; before the 2026-08-01 full-event schema change, which makes a pre-08-01 PET total a different estimator rather than a differently-trained one; and with unit measured weights and no background subtraction, which biases PET high, so the underlying gap is *larger* than the withdrawn figure. It has not been re-extracted under the pinned configuration. Because the muon is used only for selection and post-hoc binning, this is not a full joint muon–hadron unfold; ordinary closure cannot test omitted muon dependence at fixed recoil. The study tests low-level point-cloud inputs without unfolding the joint muon–recoil event. Corrected five-dimensional joint throws test the standard block-sum construction with asymmetric endpoints, a fixed estimator seed, and mean-centered throws. Those uncertainty products remain candidates until the selection-complete lateral replacement lands. The historical recoil-PET covariance is also quarantined: its fixed-map and retraining responses share nuisance endpoints and were added without their cross term. Neither a PET central-value comparison nor any PET covariance is used here, and no PET uncertainty magnitude is quoted: no PET total covariance is adopted for publication.

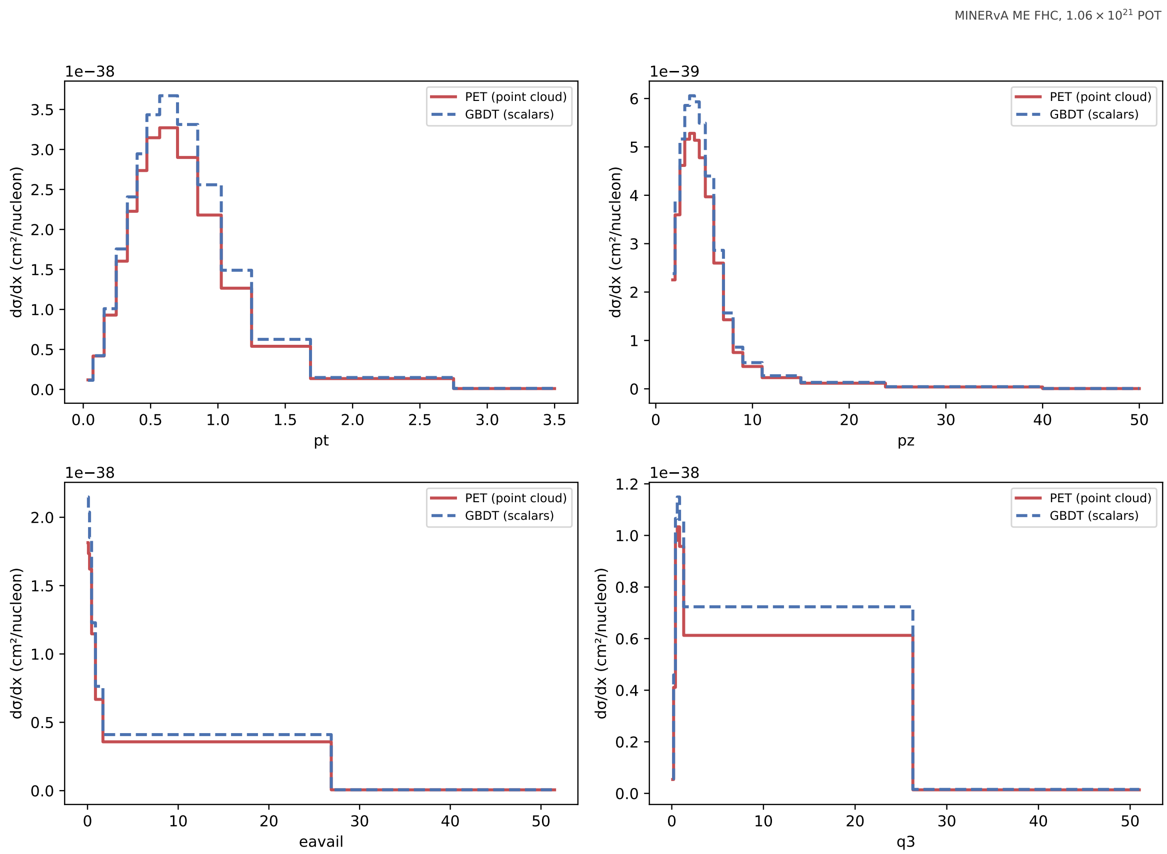

Figure 5: Recoil-only representation cross-check: absolutely-normalized 4D cross section from PET versus the production scalar GBDT, per axis. Ordinary closure was assessed only as a legacy two-iteration self-consistency diagnostic; no current closure level is quoted, and it does not test omitted muon dependence. The total offset shown is a two-iteration legacy result, withdrawn as a current comparison (see text) and not quoted here as a level of agreement.

6 Extended-fiducial central-value study

The published phase space discards about a third of the simulated signal rate (events with the muon below 1.5 GeV/c or beyond 20°); the fraction is computed from the MnvTune-weighted GENIE truth denominator and is therefore model-dependent. Because OmniFold corrects for acceptance natively, removing those truth-level cuts is a configuration change, and the unfold extrapolates into the unmeasured regions through the miss regressor. The resulting preliminary extended-fiducial CC-inclusive central value is $\sigma_{\text{ext}} = 4.502 \times 10^{-38}\,\text{cm}^2/\text{nucleon}$, 46 % above the restricted measurement (Fig. 6). Inside the published region it reproduces the restricted result to 0.57 % median per cell; a three-prior envelope reports the extrapolation cost cell by cell (few-percent where the detector has acceptance, explicit tens-of-percent flags in the dead regions). Approximately 6 % of the rate is in a newly acceptance-supported extension, while approximately 28 % is in low- or zero-efficiency cells and is separately labeled prior-dominated. The summed central value includes both tiers, but only the first is interpreted as an acceptance-supported measurement. No corrected covariance is reported for the extended domain.

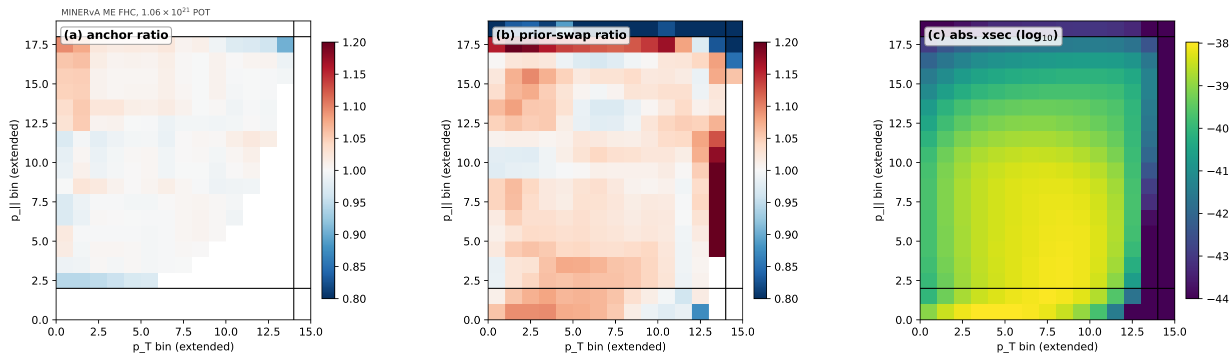

Figure 6: Extended-fiducial scalar study: (a) anchor ratio in the published sub-block (reproduces the restricted result to 0.57 %), (b) prior-swap ratio, and (c) the full cross section ($\log_{10}$). Panels (a),(b) are ratios (diverging scale centered at unity); (c) is an absolute magnitude (sequential scale).

7 Conclusions

Unbinned OmniFold reproduces a flagship MINERvA cross section and its uncertainty budget without constructing an explicit response tensor, then extends to simultaneous three-, four- and five-observable MINERvA unfolds. These independently recover the known low-recoil/2p2h deficit, agree with the dedicated published low-recoil measurement, and localize a deep-inelastic excess in a high-W corner that all tested generators underpredict. A recoil-only point-cloud transformer establishes low-level-input infrastructure without yet unfolding the complete joint muon–hadron event; that track is method development and carries no adopted uncertainty product. The scalar extended-fiducial study identifies a modest acceptance-supported extension and a much larger, separately labeled prior-dominated region.

References

[1] A. Andreassen et al. "OmniFold: A Method to Simultaneously Unfold All Observables". In: *Phys. Rev. Lett.* 124.18 (2020), p. 182001. DOI: `10.1103/PhysRevLett.124.182001`. arXiv: `1911.09107 [hep-ph]`.

[2] A. Andreassen et al. "Scaffolding Simulations with Deep Learning for High-dimensional Deconvolution". In: *ICLR simDL workshop* (2021). arXiv: `2105.04448 [stat.ML]`.

[3] D. Ruterbories et al. "Measurement of inclusive charged-current ν_μ cross sections as a function of muon kinematics at a mean neutrino energy of 6 GeV on hydrocarbon". In: *Phys. Rev. D* 104.9 (2021), p. 092007. DOI: `10.1103/PhysRevD.104.092007`. arXiv: `2106.16210 [hep-ex]`.

[4] "Measurement of ν_μ charged-current quasielastic-like cross sections in muon and proton kinematics on hydrocarbon". In: (2022). arXiv: `2203.08022 [hep-ex]`.

[5] L. Aliaga et al. "Neutrino flux predictions for the NuMI beam". In: *Phys. Rev. D* 94.9 (2016), p. 092005. DOI: `10.1103/PhysRevD.94.092005`. arXiv: `1607.00704 [hep-ex]`.

[6] "Inclusive charged-current ν_μ cross section at low momentum transfer". In: (2022). arXiv: `2110.13372 [hep-ex]`.

[7] V. Mikuni and B. Nachman. "OmniLearn: A Method to Simultaneously Facilitate All Jet Physics Tasks". In: (2024). arXiv: `2404.16091 [hep-ph]`.